Frontend Architecture

- **Prepare**
  - IDE
    - Cursor
    - Visual Studio Code
  - Terminal Emulator
    - Ghostty
    - iTerm 2
  - Shell
    - Z Shell
    - Bash
    - DevEnv
- **Scope**
  - Requirements
    - Non-Functional
    - Functional
    - User Journey
    - Accessibility (a11y)
    - Internationalization (i18n)
    - Localization (L10n)
  - Constraints
    - Compliance
    - Stakeholder Needs
- **Model**
  - User Flow
  - Data Models
    - Entities
      - Attributes

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- Schema Validator
  - Zod
- API Interfaces

Develop

- Component Patterns
  - Container/Presenter
  - Higher-Order Components
  - Hooks
- Design Patterns
  - Proxy
  - Adapter
- Linting & Formatting
  - ESlint
  - Prettier
- Architecture Styles
  - Islands Architecture
  - Micro-frontend
  - Monolith
- Repo Management
  - Turborepo
- Package Management
  - npm
  - pnpm
- Documentation Tools
  - Storybook
  - JSDoc
- Collaboration
  - GitHub
  - JIRA

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- GitLab

  Revisit GitLab offerings/capabilties

Design System

- Components
  - Unstyled primitives
    - React ARIA
    - Radix UI
  - Frameworks
    - DaisyUI
    - Chakra UI
  - Styled primitives
    - Shadcn
  - Copy/Paste
    - Aceternity UI
- Styles
  - CSS-in-JS
    - Restyle
    - StyleX

      https://stylexjs.com/
    - CSS Components

      CSS Components
  - CSS
    - Tailwind
    - Open Props

      Open Props: sub-atomic styles
    - CSS Modules
    - Frameworks
- Development
  - Storybook
- Strategy

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- Atomic Design
- Token-based Design
  - Design Tokens
    - Style Dictionary
  - Themes
    - Light/Dark Mode
- Component-driven Design (CDD)

UI Components & State

- Frameworks & Libraries
  - React
    - Next.js
  - Svelte
    - SvelteKit
  - Vue
    - Nuxt
  - Web Components
    - Lit
  - React Native
    - Expo
  - Angular
- State Management
  - Global
    - Redux
    - MobX
    - Zustand
  - Local
    - Component State
    - Context/Provider
    - Signals
    - Micro State

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- Jotai
- Zustand
- Browser
  - LocalStorage
- Async
  - Fetch state
    - isLoading/isError/isSuccess
    - Optimistic Updates
  - Caching
    - Client-side
    - In-memory
    - Browser
      - LocalStorage
  - Background/Offline
    - Service Workers
- Data Flow
  - One-way binding
  - Unidirectional
  - Observables

Data

- Fetching
  - Undici
  - window.fetch
- Modeling
  - MobX State Tree (MST)
  - Zustand Tree
- Transmission
  - REST
    - Express
    - Koa

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- Hono
- Server Sent Events (SSE)
- Realtime
  - WebSockets
  - WebRTC
- GraphQL
- Form
  - JSON Forms
    https://jsonforms.io
  - react-jsonschema-form
    react-jsonschema-form playground
- Transformation & Mapping
- Caching
  - Tanstack Query

Render & Optimize

- Build Performance
- Client-side Performance
  - Critical CSS
  - Lazy-loading
  - Code splitting
  - Hydration
  - Pre-fetching
  - Caching
  - Image Optimization
  - Core Web Vitals
- Template Engines
  - EJS
  - MDX
  - Handlebars
- Strategies

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- Client-side Rendered (CSR)
- Server-side Rendered (SSR)
- Server-side Generated (SSG)
- Progressive Web Application (PWA)
- Streaming SSR/SSG

Secure

- HTTP
  - HTTP Strict Transport Security (HSTS)
  - Cross-origin Resource Charing (CORS)
- Browser Security
  - Cross-site Scripting (XSS)
  - Cross-site Request Forgery (CSRF)
  - Input Validation
  - Content Security Policy (CSP)
- API Security
  - Authentication Mechanisms
    - Password
    - OAuth
    - Passwordless
    - Multi-factor & 2-Factor (MFA/2F)
    - Single Sign-on (SSO)
    - Token-based
  - Authentication Providers
    - AWS Cognito
    - Firebase Auth
    - Clerk
    - Auth0
  - Authorization Mechanisms
    - RBAC (Role-based)
    - ABAC (Attribute-based)

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- Roles & Capabilities
- Data Security
  - Encryption
  - Compliance

Test

- E2E
  - Playwright
  - Cypress
- Integration
- Unit
  - Vitest
  - Jest
- Mocks
- Coverage
- Testing Framework
  - Testing Library
  - Mocha

Run

- Runtime/Backend
  - Deno
  - Node

Build

- Bundle & Package
  - Vite
  - Rollup
  - Webpack
- CI Pipeline
  - GitHub Actions
- Compile
  - Turbo

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- Esbuild
- Cloud Environment

Monitor

- Observability
  - Cloudwatch
  - Datadog
  - New Relic
- Logging
  - Pino
- Metrics
  - Sentry
  - Grafana
  - Real User Monitoring (RUM)

Deploy

- CD Pipeline
  - Automation
    - Build/Test/Deploy
  - CI/CD Tools
    - GitHub Actions
- Platforms
  - Vercel
  - Deno Deploy
  - AWS
  - Hosting
    - Dynamic
      - Kubernetes
      - Docker
      - AWS Lambda
      - Serverless
    - Static

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- Vercel
- Netlify
- AWS S3
- AWS CloudFront
- Edge
  - Cloudflare Workers
  - Vercel Edge Functions
- Delivery
  - CDN
- Strategies
  - Roll-back
  - Canary
  - Blue/Green
  - A/B Test

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